Planetary Core Energy Maintenance via Solar Wind Absorption

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Abstract

We derive the energy budget for planetary core maintenance through solar wind absorption. The core receives power $P_{\text{wind}} = \eta_{\text{pen}} \cdot \Phi_{\text{sw}}(d) \cdot \pi R_{\text{mag}}^2$ from wind kinetic flux, which must balance Um maintenance cost $P_{U_m} = \mu_{\text{core}}^2 \omega_{\text{core}} / \tau_{U_m}$ and radiative losses $P_{\text{rad}} = \sigma T_{\text{core}}^4 \cdot 4\pi R_{\text{core}}^2$. The net energy rate $dE/dt = P_{\text{wind}} - P_{U_m} - P_{\text{rad}}$ determines whether a planet's core is in thermal equilibrium or depleting. The core SCm+UA is exclusively interactive with Ug3.

§1 Wind Power Absorbed by Core

$$P_{\text{wind}} = \eta_{\text{pen}} \cdot (1 - \eta_{\text{liquid}}) \cdot \frac{1}{2} \rho_{\text{sw}}(d) \cdot v_{\text{sw}}^3 \cdot \pi R_{\text{mag}}^2$$

§2 Um Maintenance Cost

$$P_{U_m} = \frac{\mu_{\text{core}}^2 \cdot \omega_{\text{core}}}{\tau_{U_m}}$$

where μ_{core} is the planetary dipole moment, ω_{core} the rotation rate, and $\tau_{U_m} \sim 10^{15}$ s the magnetic diffusion timescale.

§3 Radiative Loss

$$P_{\text{rad}} = \sigma T_{\text{core}}^4 \cdot 4\pi R_{\text{core}}^2$$

§4 Core Energy Budget

$$\frac{dE_{\text{core}}}{dt} = P_{\text{wind}} - P_{U_m} - P_{\text{rad}}$$

$$\tau_{\text{depletion}} = \frac{E_{\text{core}}}{|dE/dt|}$$

Equilibrium condition: $dE/dt \geq 0 \iff P_{\text{wind}} \geq P_{U_m} + P_{\text{rad}}$.

§5 Earth Example

Parameter	Value	Source
d	1.0 AU	Orbital distance
R_{mag}	6.4×10^7 m	Earth magnetosphere
R_{core}	3.486×10^6 m	Inner+outer core
T_{core}	5700 K	Core temperature
μ_{core}	8×10^{22} A·m^2	Geomagnetic dipole
P_{wind}	$\sim 10^{10}$ W	Wind power to core
P_{U_m}	$\sim 10^{15}$ W	Magnetic maintenance
P_{rad}	$\sim 10^{19}$ W	Radiative loss
dE/dt	< 0	Net depletion

§6 Ug3 Exclusivity

Core SCm+UA interacts exclusively with Ug3 (magnetic string rotation), not with Ug1 (dipole) or Ug2 (charge-reactivity). This means: - Core energy maintenance depends on Ug3 string coupling distance - Beyond the frost line, Ug3 coupling weakens and wind power dominates (PAPER_1079) - Core dynamo longevity correlates with Ug3 string tension retention

§7 Planetary Comparison

Planet	P_{wind}	P_{U_m}	P_{rad}	Regime
Earth	$\sim 10^{10}$ W	$\sim 10^{15}$ W	$\sim 10^{19}$ W	Depleting (slowly)
Jupiter	$\sim 10^{14}$ W	$\sim 10^{18}$ W	$\sim 10^{21}$ W	Depleting
Neptune	$\sim 10^{11}$ W	$\sim 10^{13}$ W	$\sim 10^{17}$ W	Depleting
Mars	$\sim 10^{6}$ W	~ 0 W	$\sim 10^{16}$ W	Dead dynamo

References

- PAPER_1078: Solar Wind Flux Partitioning Budget
- PAPER_1079: Frozen Planet Solar Wind Power
- PAPER_413: Ug3CCWCWDifferentialRotationSCmPlanetaryCoreCalculator
- Star-Magic.txt: §5 Core Energy Maintenance

Key References with arXiv/DOI Identifiers

1. Abbott et al. (LIGO Scientific and Virgo Collaborations, 2016). *Observation of Gravitational Waves from a Binary Black Hole Merger.* Phys. Rev. Lett. **116**, 061102 — arXiv:1602.03837 — doi:10.1103/PhysRevLett.116.061102
2. Murphy, D. (2026). *Unified Quantum Field Framework (UQFF): Star-Magic v5.x Whitepaper Series.* Star-Magic Repository — github.com/Daniel8Murphy0007/Star-Magic